# Day 3 introduction IP

Task 1:- List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

## Differences Between IPv4 and IPv6

| IPv4 | IPv6 |
|---|---|
| Uses 32-bit addressing. | Uses 128-bit addressing. |
| Written in decimal format. | Written in hexadecimal format. |
| Example: `192.168.1.1` | Example: `2001:db8::1` |
| Supports approximately 4.3 billion addresses. | Supports a very large number of addresses. |
| Uses dotted decimal notation. | Uses colon-separated notation. |
| Supports broadcast. | Does not support broadcast; uses multicast instead. |
| NAT is commonly used. | Generally does not require NAT. |
| Header size is variable (20–60 bytes). | Header size is fixed (40 bytes). |

Task 2:- Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

### Static IPv4 Address Assignment (192.168.10.0/24)

| Device | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| Laptop | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 |
| Phone | 192.168.10.3 | 255.255.255.0 | 192.168.10.1 |
| Smart TV | 192.168.10.4 | 255.255.255.0 | 192.168.10.1 |
| Printer | 192.168.10.5 | 255.255.255.0 | 192.168.10.1 |

Network Details:

- Network Address: `192.168.10.0`
- Default Gateway: `192.168.10.1`
- Subnet Mask: `255.255.255.0`
- Broadcast Address: `192.168.10.255`

Task 3:- Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

## Given:

- Network: `192.168.1.0/24`
- Subnet Mask: `/27` = `255.255.255.224`
- Borrowed bits: 3
- Host bits: 5

## 1. Calculate Subnets and Hosts

- Number of subnets = \(2^3 = 8\)
- Hosts per subnet = \(2^5 - 2 = 30\)
- Valid usable hosts = 30 per subnet

## 2. First Three Subnet Ranges

| Subnet | Network Address | Usable Host IP Range | Broadcast Address |
|---|---|---|---|
| 1 | 192.168.1.0 | 192.168.1.1 – 192.168.1.30 | 192.168.1.31 |
| 2 | 192.168.1.32 | 192.168.1.33 – 192.168.1.62 | 192.168.1.63 |
| 3 | 192.168.1.64 | 192.168.1.65 – 192.168.1.94 | 192.168.1.95 |

## 3. Final Answer

- Total Subnets: 8
- Total Addresses per Subnet: 32
- Usable Hosts per Subnet: 30
- Subnet Increment: 32

Conclusion: A `/27` subnet divides the `192.168.1.0/24` network into 8 subnets, with 30 usable host addresses in each subnet.

Task 4:- You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. the subnet mask to CIDR notation and use bitwise AND for network address calculation.

- P Address: `10.0.5.17`
- Subnet Mask: `255.255.255.0`

### 1. Convert Subnet Mask to CIDR

`255.255.255.0` = /24

### 2. Network Address (Bitwise AND)

```
IP Address:     10.0.5.17
Subnet Mask:    255.255.255.0
--------------------------------
Network:        10.0.5.0
```

Bitwise AND operation gives the Network Address: 10.0.5.0

### 3. Broadcast Address

`10.0.5.255`

### 4. Valid Host Range

- First Host: `10.0.5.1`
- Last Host: `10.0.5.254`

### Final Answer

| Details | Result |
|---|---|
| CIDR Notation | /24 |
| Network Address | 10.0.5.0 |
| Broadcast Address | 10.0.5.255 |
| Valid Host Range | 10.0.5.1 – 10.0.5.254 |
| Total Usable Hosts | 254 |

Task 5:- Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

## ubnet Calculator Verification

Step 1: Open the online subnet calculator:

Online Subnet Calculator

Step 2: Enter these values:

- IP Address: `10.0.5.17`
- Subnet Mask: `255.255.255.0`

Step 3: Expected Calculator Output

| Field | Calculator Result |
|---|---|
| CIDR | /24 |
| Network Address | 10.0.5.0 |
| Broadcast Address | 10.0.5.255 |
| First Usable IP | 10.0.5.1 |
| Last Usable IP | 10.0.5.254 |
| Usable Hosts | 254 |